Manuel Savenia

Computer Engineering Student focused on Embedded Systems and ML

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Portfolio: manusave.engineer

PROFESSIONAL PROFILE

Advanced Computer Engineering student with a strong interest in the convergence of hardware and software. My primary focus is on **Embedded Systems** development and the optimization of **Machine Learning** algorithms for **Edge Computing** devices. I have hands-on experience designing custom PCBs, programming firmware in C, and deploying distributed architectures using containers and microservices. I stand out for my ability to solve complex technical problems and my adaptability to work in collaborative environments under **Agile methodologies**.

EDUCATION

Computer Engineering 2022 – Present Universidad Nacional de La Plata
Language School 2012 – 2020 Universidad Nacional de La Plata

PROFESSIONAL EXPERIENCE

PC Repair Technician **Time Travellers**
2023 – 2026 *La Plata, Arg.*

- Advanced diagnosis and preventive maintenance of high-complexity hardware.
- Operating system performance optimization and electronic repair at the component level (SMD).

TECHNICAL SKILLS

- **Programming Languages**: Python (TensorFlow, FastAPI), Java, C, SQL, Assembly (x86), MATLAB, Pascal.
- **Engineering & Hardware**: PCB Design (KiCad, Proteus), Analog/Digital Electronics, Microcontrollers (ESP32, Arduino).
- **Frontend**: React, Next.js, Tailwind CSS, Node.js.
- **Tools & Systems**: Git, Docker, Linux (Arch, Debian), Figma, Jupyter Notebooks.

EXPERIENCE AND PROJECTS

ElectricSim - Electrical Demand Prediction **2025**
Rol: ML Developer & Data Engineer (Python, TensorFlow)

- Developed and trained a Deep Learning model using TensorFlow to predict electrical demand in Greater Buenos Aires, focusing on large-scale data preprocessing and model accuracy optimization.

Repository: github.com/ElectricSIM/ElectricSim-ML

Blitzcrank - Mobile Robotic Arm **2024**
Rol: Hardware & Firmware Lead (C, KiCad, PCB Design)

- Led the development of the electrical infrastructure and control firmware, designing a **custom PCB** for efficient power management and coordinated servomotor handling.

Repository: github.com/gonblas/blitzcrank

Biped Robot - Bipedal Structure and Control **2024**
Rol: Control Systems Developer (C, ESP8266, Arduino)

- Built the physical structure for a bipedal prototype, developing a **web telemetry interface** on ESP8266 for live movement parameter adjustments.

Information: manusave.engineer/projects/bipedal-robot

Room LEDs - Smart Lighting (WLED) **2024**
Rol: Firmware & IoT Integrator (ESP32, WLED)

- Implemented a home automation lighting system synced with audio, configuring **WLED** firmware on ESP32 microcontrollers and customizing reactive effects.

Information: manusave.engineer/projects/room-leds

CERTIFICATIONS AND OTHERS

- **EF SET English Certificate**: C1 Advanced Level (2025).
- **Introduction to IoT and Digital Transformation**: Cisco Network Academy (2025): credly.com/badges.
- **Languages**: Spanish (Native), English (C1): cert.efset.org.